

System and Network Security

Dr. Ashok Kumar Das

IEEE Senior Member
Professor

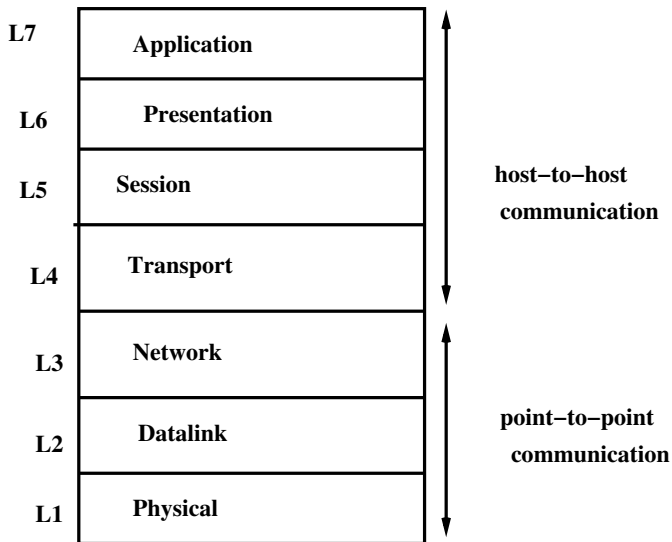
Center for Security, Theory and Algorithmic Research
International Institute of Information Technology, Hyderabad

E-mail: *ashok.das@iiit.ac.in*

URL: <https://www.iiit.ac.in/faculty/ashok-kumar-das/>
<https://sites.google.com/view/iitkgpakdas/>

- In 1978, International Standards Organization (OSI) proposed a 7-layer reference model for network services and protocols, known as the OSI model.
- The main objective of the OSI model as
 - (1) Systematic approach to design.
 - (2) Changes in one layer should not require changes in other layers.

The OSI Reference Model



Physical Layer:

- Transmits raw bit stream over a physical medium.
- The design issues have to do making sure that when one side sends a 1 bit, it is received by the other side as a 1 bit, not a 0 bit.
- The design issues largely deal with mechanical, electrical, and timing interfaces, and the physical transmission medium, which lies below the physical layer.
- Network components: Repeater, Multiplexer, Hubs, Amplifier.

Datalink Layer:

- Reliable transfer of frames (data) over a point-to-point link.
- Responsible for flow control, error control (error detection/correction), congestion control.
- Network components: Bridge, Switch, NIC, Advanced Cable Tester.

Network Layer:

- Establishing, maintaining and terminating connections.
- Routes packets (messages) through point-to-point link.
- Network components: Router, Frame Relay Device, ATM Switch.

Transport Layer:

- End-to-end reliable data transfer, with error recovery and flow control.
- Network components: Gateway.

Session Layer:

- Allows users on different machines (hosts) to establish sessions between them.
- Session offer various services, including
 - ▶ Dialog Control: Keeping track of whose turn it is to transmit.
 - ▶ Token Management: Preventing two parties from attempting the same critical operation at the same time.
 - ▶ Synchronization: Checkpointing long transmissions to allow them to continue from where they were after a crash.
- Network components: Gateway.

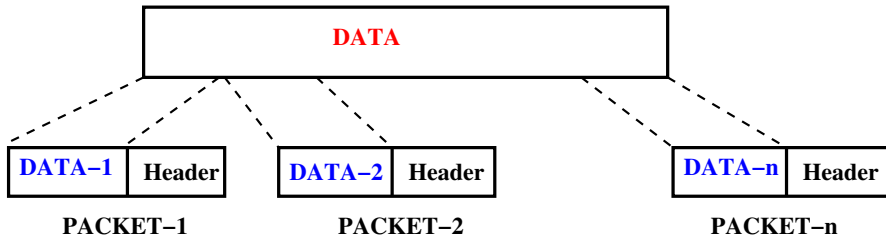
Presentation Layer:

- Translates data from application to network format, and vice-versa.
- All different formats from all sources are made into a common uniform format that the rest of the OSI model can understand.
- Network components: Gateway.

Application Layer:

- Interface point for user applications.
- Network components: Gateway.

Data handled in a particular layer:



Thank You!!!